SABRINA WILLIAMS

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PROFESSIONAL SUMMARY

Data Analyst with a strong foundation in clinical and biomedical domains. Experienced in programming (MATLAB, Python, SQL), clinical data workflows, medical technologies, and interdisciplinary research. International academic background, fast learner, and passionate about data-driven healthcare innovation.

TECHNICAL SKILLS

- **Programming**: MATLAB, Python, SQL, R, C, Java, JavaScript, HTML/CSS
- **Data Tools**: Statistical analysis, data merging, data cleaning, preprocessing, visualization, UI development
- **Clinical Tools**: experiment design, motion capture, force platforms, accelerometers, IMUs, exosuits, MRI data handling
- **Machine Learning:** Supervised and unsupervised ML, classification, regression
- **Software**: Microsoft Office, GitHub, online database systems

RELEVANT EXPERIENCE

Master's Thesis Researcher

Harvard University – School of Engineering and Applied Sciences, Biodesign Lab, *Boston, MA*
Jun 2025 – Dec 2025 · Exchange Semester

Master's Thesis: Development of a Standardized Cross-Study Gait Dataset and Exploratory Analysis of Post-Stroke Walking Patterns

- Built a MATLAB data pipeline to merge 8 independent gait studies into a standardized dataset with ~60.000 gait cycles from 29 post-stroke participants
- Performed large-scale data cleaning, preprocessing, synchronization, and quality control to enable analysis-ready biomechanical datasets for rehabilitation research

AI Developer

Wallinger Ricker Schlotter Tostmann, *Munich, Germany*
Apr 2025 – May 2025

- Designed an internal AI system to process and retrieve technical patent data using embeddings and LLMs
- Improved internal document search and access with semantic query support

Data Analysis Tutor

Justus-Liebig-University, *Giessen, Germany*
Apr 2024 – Aug 2024 · Voluntary

- Tutored students in applied data analysis concepts and methodologies
- Communicated complex analytical tropics in a clear, application-oriented manner

Programming Tutor (MATLAB)

Justus-Liebig-University, *Giessen, Germany*
Jun 2024 – Jul 2024

- Delivered intensive MATLAB crash course focused on data handling, preprocessing, and GUI development
- Developed hands-on teaching content and structured lessons for quick learning

Graduate Student Research Assistant (Cognitive Neuroscience)
Justus-Liebig-University, *Giessen, Germany*
Apr 2024 – Jun 2024

- Supported MRI-based experimental studies, including data acquisition and experiment setup
- Assisted with research workflows, data quality control, and test subject management

Undergraduate Research Assistant
University of Regensburg, *Regensburg, Germany*
Apr 2022 – Apr 2023

- Conducted literature reviews, experiment planning, lab work, and research documentation
- Assisted in editing and preparing publications for journal submission

Team Lead – COVID-19 Testing Station
21Dx GmbH, *Munich, Germany*
Oct 2020 – Dec 2021

- Managed site operations, safety regulations, scheduling, and financials during peak testing periods
- Supervised and trained a rotating staff team and ensured efficient onboarding of new personnel

EDUCATION

Justus-Liebig-University, Giessen, Germany
Oct 2023 – Jan 2026

- **M.Sc. in Human Movement Analytics, overall GPA 3.8**
- Key subjects: Mathematics, Statistics, Medical Engineering, Biomechanics, Neuroscience, Experiment Design, and Statistical Modelling

Study Abroad, Università di Torino, Turin, Italy
Oct 2023 – Jan 2026

- Completed course work in Biomechanics of Movement, Clinical Neuroimaging, Programming for Data Science, Neuroimaging Technologies, and Statistical Methods

University of Regensburg, Regensburg, Germany
Jul 2023

- **B.A. in Applied Movement Science, overall GPA 3.7**
- Minor: Information Science

CERTIFICATIONS & TRAINING

- Supervised Machine Learning (Stanford University via DeepLearning.AI)
- Advanced Learning Algorithms (Stanford University via DeepLearning.AI)
- Biomedical Research Investigators (CITI Program)
- General & Laboratory Safety, Laser & Soldering Safety (Harvard EHS)
- Introduction to Clinical Data (Stanford University, School of Medicine)
- SQL for Data Science (University of California, Davis)
- Introduction to Healthcare (Stanford University, School of Medicine)

LANGUAGES

- German – Native
- English – Fluent
- French – Conversational
- Italian – Basic